NMC 2014 O-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn. 1. Given that $k \div 2 \div 2 \div 2 \div 2$ and $n \div 2 \div 2 \div 2$ are each equal to the smallest positive integer. Find the value of $k + n$.

Qn. 2. Simplify the following expression $(2014)^2 - (2)(2014)(2013) + (2013)^2$.

Qn. 3. The average of the digits of the integer 46 is 5. Including 46, how many two-digit positive integers have the average of their digits equal to 5?

Qn. 4. A child's age, increased by 3, gives a perfect square, and when decreased by 3 the age is the square root of that perfect square. How old is the child?

Qn. 5. When counting from 3 to 531, 53 is the 51st number counted. When counting backwards from 531 to 3, 126 is the nth number counted. What is n?

Qn. 6. A bag contains 65 balls of the same size. There are 20 red ones, 20 green ones, 20 blue ones and another 5 that are either yellow or white. Lorinda removes balls from the bag without looking. What is the smallest number of balls that she must remove to ensure that she has 10 of the same colour?

Qn. 7. Tacia participates at a free throw basketball contest. For each successful throw she makes, she gets 8 points and for each miss she is penalized 5 points. After 26 throws she has 0 points. How many successful throws did she make?

Qn. 8. Mary paid shs 480 to purchase a certain number of items, but the nice vendor gave her two extra. This decreased the price per item by shs 1. How many items did she receive (including the two extra)?

Qn. 9. Consider the positive integer p with representation $p = 73961w$. If p divided by 5 has a remainder of 4, and p divided by 3 has a remainder of 0, what is the value of w?

Qn.10. In order to estimate the value of $x - y$ where x and y are real numbers with $x > y > 0$, Chantal rounded x up by a small amount, rounded y down by the same amount and then subtracted her rounded values. Which of the following statements is necessarily correct? Show your working.
(A) Her estimate is larger than $x - y$. (B) Her estimate is smaller than $x - y$.
(C) Her estimate equals $x - y$. (D) Her estimate equals $y - x$. (E) Her estimate is 0.

SECTION B

Qn.11. When a decimal point is placed between the digits of the two-digit whole number n, the resulting number is equal to the average of the digits of n. What is the value of n?

Qn.12. Each of Alice, Bob, Carol and Don took a test. Each of them answered atleast one question correctly and altogether they answered 67 questions correctly. Alice had more correct answers than anyone else. Bob and carol together answered 43 questions correctly. How many correct answers did Don have?

Qn.13. Miranda bought some toy bicycles and some toy helicopters. She paid a total of $shs201$. She did not buy partial toy bicycles or partial toy helicopters. Each toy bicycle cost $shs19$ and each toy helicopter cost $shs17$. How many of each did she buy?

Qn.14. How many numbers between 10 and 99 have digits that differ by exactly 4?

Qn.15. In the diagram, the rectangle is divided into nine smaller rectangles. The areas of five of these rectangles are given. Determine the area of the rectangle labelled R.

3	1	
	2	R
5		10

NMC 2014 O-LEVEL PAPER 2 SOLUTIONS

SECTION A

Qn1. $k \div 2 \div 2 \div 2 \div 2 = 1 \Longrightarrow k \div 16 = 1$ so that $k = 16$. Also $n \div 2 \div 2 \div 2 = 1 \Longrightarrow n \div 8 = 1$ so that $k = 8$. Therefore $k + n = 16 + 8 = 24$.

Qn2. $(2014)^2 - (2)(2014)(2013) + (2013)^2 = (2014 - 2013)^2 = 1^2 = 1$.

Qn3. For the average of two digits to be 5, their sum must be 10. The two-digit positive integers whose digits sum to 10 are $19, 28, 37, 46, 55, 64, 73, 82, 91$ of which there are 9.

Qn4. If the child's age is x, then $x+3 = (x-3)^2$, which simplifies to $x^2-7x+6 = 0$ or $(x-1)(x-6) = 0$, so $x = 1$ or $x = 6$. However, if $x = 1$, then $x - 3 = -2$, which is not the square root of 4, since the child's must be positive. Thus $x = 6$ implying that the child's age is 6 years.

Alternatively,

By trial and error, the age is 3 less than a perfect square, so the only possibilities for a child are 1 or 6 or 13 and 6 is the only one that satisfies the other condition.

Qn5. The number 531 is the 1st number counted when proceeding backwards from 531 to 3. In turn, 530 is the 2nd number, 529 is the 3rd number and x is the $(531 - x + 1)^{th}$ number. Therefore 126 is the $(532 - 126)^{th}$ number, which is the 406th number.

Qn6. To avoid having ten balls of the same colour, Tacia can remove up to nine of each of the colours red, green, blue, plus the five other balls, making a maximum total of 32 balls. Thus in order to ensure ten of the same colour, she needs to remove $32 + 1 = 33$ balls.

Qn7. Let b represent baskets and m represent missed. Then $8b - 5m = 0$ and $b + m = 26$. Solving simultaneously for b and m gives $b = 10$. Thus he had 10 successful throws.

Qn8. Let x be the number of items that he received, then

$$\frac{480}{x-2} - \frac{480}{x} = 1$$

This simplifies to $960 = x(x-2)$, yielding $x = 1 + \sqrt{961} = 1 + 31 = 32$.

Alternatively

Let x be the number of items that she would have received if the vendor had not given her extra two items, then

$$\frac{480}{x} - \frac{480}{x+2} = 1$$

This simplifies to $960 = x(x+2)$, yielding $x = \frac{-2 \pm \sqrt{3844}}{2} = -2 \pm 32 = 30$ since the negative value is ignored as she did not buy negative items. Therefore including the two items, Mary received $x + 2 = 30 + 2 = 32$ items.

Qn9. $73961a$ is divisible by 3 if $7+3+9+6+1+w$ is a multiple of 3. This implies that the possible values of w are $1, 4$ or 7. Since $73961w$ divided by 5 has a remainder of 4, then w is either $0+4=4$ or $5+4=9$. Since 739619 is not a multiple of 3, it implies w must be 4.

Qn10. Consider x and y as points on the real number line, with x necessarily to the right of y. Then $x-y$ is the distance between x and y. Chantal's rounding moved x to the right and moved y to the left. Therefore the distance between them increased, and her estimate is larger than $x-y$.

To see that the other answer choices are not correct, let $x=2.9$ and $y=2.1$, and round each by 0.1. Then $x-y=0.8$ and Chantal's estimated difference is $(2.9+0.1)-(2.1-0.1)=1.0$.

SECTION B

Qn11. Suppose that n has digits AB. Then $n=10A+B$. The average of the digits of n is $\frac{A+B}{2}$. Putting a decimal point between the digits of n is equivalent to dividing n by 10, so the resulting number is $\frac{10A+B}{10}$.
So we want to determine A and B so that

$$\frac{A+B}{2}=\frac{10A+B}{10}$$

$$10A+B=5(A+B)$$
$$5A=4B$$

Since A and B are digits such that $5A=4B$, then $A=4$ and $B=5$ is the only possibility. Therefore, $n=45$. (We can quickly check that the average of the digits of n is 4.5, the number obtained by putting a decimal point between the digits of 45.)

Alternatively,

When we compute the average of two digits, the result is either an integer or a half-integer (that is a decimal number of the form $x.5$). Therefore, the possible averages are $0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0$. (Note that 0.0 is not possible since the smallest two-digit whole number is 10 and the average of digits 1 and 0 which gives the smallest average of the digits is 0.5). From this list, the only one equal to the average of the two digits forming it is 4.5. Therefore $n=45$ (formed by removing the decimal point from 4.5).

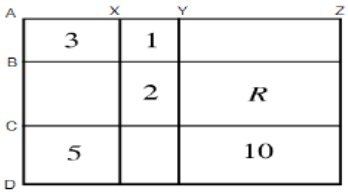
Qn12. Since one of Bob and Carol had atleast 22 questions answered correctly, Alice must have had atleast 23. Thus, the three of them together had atleast $43+23=66$ questions answered correctly. Thus Don answered atmost $67-66=1$ and since everyone had exactly one correct answer, he had exactly one correct answer.

Qn13. Suppose that Fiona bought x toy bicycles and y toy helicopters. Then $19x+17y=201$. Since x and y are non-negative integers, then $19x \leq 201$ so $x \leq 10$.
Suppose $x=10$, then $17y=201-19x=11$, which does not have an integer solution because 11 is not divisible by 17.

Suppose $x = 9$, then $17y = 201 - 19x = 30$, which does not have an integer solution.
Suppose $x = 8$, then $17y = 201 - 19x = 49$, which does not have an integer solution.
Suppose $x = 7$, then $17y = 201 - 19x = 68$, so $y = 4$. Therefore, $19(7) + 17(4) = 201$, and so Fiona buys 7 toy bicycles and 4 toy helicopters.
(We can check that $x = 0, 1, 2, 3, 4, 5, 6$ do not give values of y that work.)

Qn14. Let the tens digit be t and the units digit be u, so $t - u = \pm 4$. Also $1 \leq t \leq 9$ and $0 \leq u \leq 9$. (Note that the tens digit cannot be zero.) If $t = u + 4$, then $0 \leq u \leq 5$ which gives six numbers. If $t = u - 4$, then $5 \leq u \leq 9$, giving another five numbers, for a total of 11.

Qn15. Looking at the middle column of rectangles, each has the same width, so the ratio of their areas equals the ratio of their heights. Thus, $AB : BC = 1 : 2$. Looking at the rectangles in the first column, the area of the middle rectangle must be twice the area of the top rectangle, or 6. Thus, $BC : CD = 6 : 5$ by the reasoning above. So, looking at the third column, $R : 10 = 6 : 5$ or $R = 12$.

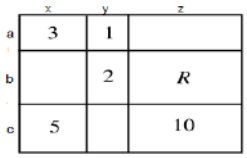


Alternatively

Let the width of the first column be x.
Since the area of the top left rectangle is 3, the height of the first row is $\frac{3}{x}$.
Since the area of the bottom left rectangle is 5, the height of the third row is $\frac{5}{x}$.
Since the height of the first row is $\frac{3}{x}$ and the area of the top middle rectangle is 1, the width of the middle column is $\frac{3}{x}$. Thus, the height of the middle row is $\frac{6}{x}$, since the area of the middle rectangle is 2.
Since the height of the third row is $\frac{5}{x}$ and the area of the bottom right rectangle is 10, then the width of the third column is $2x$.
Since the rectangle labelled R has height $\frac{6}{x}$ and width $2x$, then it has area 12.

Alternatively

We label some of the lengths in the diagram.



From the given information, $ax = 3, ay = 1, by = 2, bz = R, cx = 5$ and $cz = 10$. We want to determine bz. But $bz = \frac{(ax)(by)(cz)}{(ay)(cx)} = \frac{(3)(2)(10)}{(1)(5)} = 12$, so $R = 12$.